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IEEE TWC- Decision on Manuscript ID Paper-TW-Apr-26-1172

1 message

IEEE Transactions on Wireless Communications <onbehalfof@manuscriptcentral.com> Sun, Aug 23, 2026 at 9:11 AM

Reply-To: shuowen.zhang@polyu.edu.hk
To: l.venturino@unicas.it
Cc: yz4891@columbia.edu, l.venturino@unicas.it, xw2008@columbia.edu, c.masouros@ucl.ac.uk, shuowen.zhang@polyu.edu.hk, eic.twc@ece.ubc.ca, admin.twc@ece.ubc.ca

23-Aug-2026

*** Please submit your revision in double-column. The detailed format requirements can be found on the TWC website (https://www.comsoc.org/publications/journals/ieee-twc/submit-manuscript).

Dear Prof. Venturino:

Manuscript ID Paper-TW-Apr-26-1172, titled "Direct and Ambient Backscatter Communications with a Dual-Function Radar Transmitter" and submitted to the IEEE Transactions on Wireless Communications, has been reviewed. The comments of the reviewer(s) are included at the bottom of this letter.

The reviewer(s) have suggested some major revisions to your manuscript. Therefore, I invite you to respond to the reviewer(s)' comments and revise your manuscript.

Please provide here your detailed editorial summary that explains and justifies your decision.

The reviewers have raised some major concerns. Based on the reviewers' comments and my own reading, a major revision is suggested for this round. The main concerns include the following:

1. The authors are suggested to compare their proposed scheme with prior studies from other groups, as pointed out by Reviewer 1. The actual radar sensing performance should also be evaluated, as pointed out by Reviewer 2 and Reviewer 3.

2. The Doppler effect needs to be discussed especially since this paper makes use of the periodicity of radar pulses. Some other practical issues are also suggested to be discussed as pointed out by multiple reviewers.

3. The perfect timing synchronization assumption and the sparse channel assumption are suggested to be further justified and discussed, as pointed out by Reviewer 2.

4. The actual computation time is suggested to be evaluated, as pointed out by Reviewer 3.

5. There are also other concerns regarding the technical details, presentation, notation, etc.

It should be emphasized that a major revision decision does not imply eventual acceptance. A future version of the paper will only be accepted if the revision addresses the major concerns, and the reply is highly persuasive to the Editors and Reviewers.

Once the revised manuscript and a point-by-point reply to the reviewer comments is prepared, you can upload them both and submit through your Corresponding Author Center. The reply should be designated as a "supplementary file.

When you are ready to submit your revision, visit the following link:
https://ieee.atyponrex.com/submission/submissionBoard/REX-PROD-2-A8AE019E-D075-4EA2-A55E-910A02A2894A-DC58475D-8BF4-43F9-85C1-4903FEB63FC6-77313/current?idtype=external

IMPORTANT: Your original files are available to you when you upload your revised manuscript. Please delete any redundant files before completing the submission.

When you revise your manuscript, please make sure that you comply with our manuscript length requirements. Your revised manuscript should be returned within EIGHT WEEKS. We will consider the manuscript as rejected if the revised manuscript is not received within EIGHT WEEKS from the date of this letter.

Once again, thank you for submitting your manuscript to the IEEE Transactions on Wireless Communications and I look forward to your revision.

Sincerely,

Prof. Shuowen Zhang
Editor, IEEE Transactions on Wireless Communications
shuowen.zhang@polyu.edu.hk

Reviewer(s)' Comments to Author:
Reviewer: 1

Comments to the Corresponding Author
This paper investigates the joint design of direct communication and ambient backscatter communication within a dual-function radar communication system. Two signaling schemes (pilot-free and pilot-aided) are proposed, along with corresponding joint and non-joint decoding algorithms. Overall, the topic aligns well with current trends in ISAC and green IoT. The reviewer has following concerns:
1.In Fig. 6, the proposed scheme is compared with references [30] and [32]. References [30] and [32] include some of the same authors as this submission. Please include comparisons with recent independent works, such as Ref. [16] (Y. Tian et al., 2024) or Ref. [33] (S. Zeng et al., 2025). Ref. [33], in particular, also investigates Radar-enabled ISAC systems. Comparing against such independent works would significantly strengthen the paper's claims.
2.The manuscript assumes the channel remains constant within a frame (Assumption A2) and exploits the periodicity of radar pulses. Radar systems are frequently deployed to detect high-speed moving targets (e.g., vehicles). If the target or Tag moves at high speeds, the Doppler shift will disrupt the coherent cumulation between pulses, leading to degraded demodulation performance for x_p .
3.Please check the notation throughout the manuscript. For example, near Eq. (14), the dimensions of y are defined. It would be helpful to explicitly clarify how the specific values of L (number of PRIs) and K (fast-time samples) affect the condition number of the matrix
4.In abstract: "...enables joint direct andbackscatter..." is missing a space. It should be "and backscatter".
5.Page 1, Right Column: "Difference: Unlike these prior works..." This text appears to be a residual artifact from the submission system's metadata. Please remove or reformat it in the final version.
6.The definition of the transmission rate seems redundant.

Reviewer: 2

Comments to the Corresponding Author
Please find my comments below:

1.The perfect timing synchronization assumptions is impractical and the robustness analysis is required. The whole signal matrix \mathbf{Y} and subspace separation logic rely on Assumption A1, which only guarantees coarse PRI/frame synchronization via beacons but ignores residual fine timing errors. Passive tags only adopt low-precision RC oscillators, which will produce accumulated clock drift within one frame. Small chip-level time shifts break the regular fast-slow matrix structure and destroy the orthogonality between \boldsymbol{x} and \mathbf{1}_L. The manuscript merely claims timing offsets can be absorbed into channel vectors without any quantitative simulation or theoretical bound to prove the decoder's tolerance range of time drift. Compared with OFDM-based cooperative ambient backscatter that uses cyclic prefix to resist minor misalignment, this pulse-frame DFRC system imposes stricter synchronization requirements, yet the paper never highlights this drawback or provides any drift compensation scheme. Simulations with varying timing offsets should be supplemented to verify real-world robustness. Please note that existing works have shown that the perfect is impossible.
[1]Y. Feng, S. Chen, W. Xi, S. Wang, J. Zhao, and W. Gong, "Heartbeating with LTE networks for ambient backscatter," IEEE Trans. Mobile Comput., vol. 23, no. 5, pp. 4246–4258, 2024
[2]Y. Ye, Y. Li, X. Chu, G. Zheng and S. Sun, "Symbol Detection in Ambient Backscatter Communications Under Residual Time Synchronization Errors," IEEE Trans. Commun., vol. 74, pp. 1669-1685, 2026
2.The paper states the tag code constraint \boldsymbol{x}^H\mathbf{1}_L=0 as the core condition for perfect signal separation, but it fails to distinguish transmit-side code orthogonality and effective orthogonality after channel compensation. Two confusing points exist: First, the raw received signal \mathbf{Y} = \boldsymbol{x}\boldsymbol{\alpha}_{STR}^T + \mathbf{1}_L\boldsymbol{\alpha}_{SR}^T mixes unknown channel vectors, so \boldsymbol{x} and \mathbf{1}_L are not orthogonal on the receiving side before channel estimation. Orthogonality only works after we estimate and remove the fast-time channel weight \boldsymbol{\alpha}_{STR}/\boldsymbol{\alpha}_{SR}. Second, projecting received data onto the subspace spanned by \boldsymbol{x} requires prior knowledge of \boldsymbol{x}. The paper does not clearly clarify that blind traversal over all candidate tag codewords is mandatory before using the orthogonal projection property. Readers may mistakenly believe the two signal components can be directly separated from raw received samples without decoding. The authors need to rewrite relevant theoretical sections to clarify this precondition and add intuitive mathematical derivations.
3.The manuscript targets a dual-function radar-communication transmitter, meaning radar ranging and target

detection are core system functions alongside communication. However, all simulation results only evaluate source/tag BER and channel NRMSE, with no metrics related to radar sensing performance. The paper cannot demonstrate whether introducing tag backscatter and the proposed joint decoder will degrade radar sensing accuracy, which severely weakens the integrity of its DFRC research.
4.Just a discussion. My group has implemented real-world ambient backscatter tests using USRP hardware. The measured sampled data clearly demonstrates non-uniform ADC sampling intervals and occasional sample loss during signal reception. So I guess that even though the tag transmits short frames with a small number of symbols, slight sampling irregularities still distort the fast-time channel vectors and introduce inter-PRI signal leakage, partially breaking the orthogonality between \boldsymbol{x} and \mathbf{1}_L. For long frames with numerous tag symbols, cumulative sampling jitter and missing samples will completely destroy the regular two-dimensional matrix structure and render the proposed joint/disjoint decoders invalid. How to deal with this issue?
5.The advantage of l1-regularization shown in Fig. 3 relies on the assumption of sparse channels. However, in rich scattering environments (e.g., urban canyons), channels can be dense. Please add some simulations
6.Lastly, in eq. (14), the notation \sum is used for the signal matrix. Please ensure this does not cause confusion with summation operators in the text.

Reviewer: 3

Comments to the Corresponding Author
This paper presents a well-structured framework for simultaneous direct and ambient backscatter communications using a dual-function radar transmitter. The fast-time/slow-time subspace formulation is interesting, and the paper provides comprehensive analyses. Overall, the manuscript is technically solid. The following points would further strengthen the paper.
1. The proposed model assumes frame-level channel invariance, accurate synchronization, and known delay bounds. Please evaluate or discuss the robustness of the proposed decoders to residual Doppler/CFO, timing offsets, and delay-range mismatch.
2. Since the source is a dual-function radar transmitter, the sensing aspect could be evaluated more thoroughly. Besides the reported PSL and ISLR results, it would be helpful to show how embedding communication data affects representative radar detection or range-estimation performance.
3. Although asymptotic complexity expressions are provided, the practical computational cost remains unclear. Please report representative runtime or operation-count comparisons among joint, disjoint, non-iterative, and iterative decoding, particularly as the codebook sizes and channel delay spread increase.
4. Please avoid typos. For example, the sentence “the relaxed data symbol updates are now updated as follows” is repetitive and could be revised to “the relaxed data-symbol updates are performed as follows.